

Hardware Business Management System

Purpose of This Document

Core entities, data rules, database planning notes, and accounting decisions required before final schema. This standalone file is written for handoff to a Cloud AI, software developer, designer, database planner, or project reviewer.

How to Use This Document

- Use Confirmed requirements as agreed scope for planning and implementation.
- Use Proposed requirements as recommended starting points that can still be adjusted.
- Use Pending Decision items as questions for the business owner before coding or final schema design.
- Do not treat unresolved accounting, profit, allocation, stock costing, tax, or debt rules as final until the owner confirms them.

Status Label Convention

Confirmed means the item is part of the agreed scope. Proposed means it is a recommended implementation detail that can still change. Pending Decision means the business owner must decide before a developer or Cloud AI treats it as final.

Status	Meaning	Developer Handling
Confirmed	Part of the agreed project scope.	Implement unless later changed by the owner.
Proposed	Recommended approach or useful feature.	Validate during design and planning.
Pending Decision	Business rule or accounting detail not fully decided.	Ask the owner or make configurable. Do not hard code.

Document 05 Data Model and Database Requirements

This section lists core data entities and relationships for implementation planning. It is not a final physical schema. Developers should turn it into a normalized database design, migration files, API contracts, and SQLite sync tables.

Core Entities

Entity	Purpose	Important Fields	Status
Business	Stores owner business profile.	name, contact, currency, settings	Confirmed
Shop	Represents each hardware shop or branch.	business id, shop name, location, active status	Confirmed
User	Stores system users.	name, phone or email, password hash, role, status	Confirmed
Role Permission	Controls allowed features.	role, permission code, allowed flag	Confirmed
Product Category	Groups products.	category name, active status	Confirmed
Product	Stores hardware product information.	name, code, unit, category, cost, selling price, reorder level, status	Confirmed
Stock Balance	Stores current quantity by shop and product.	shop id, product id, quantity, updated at	Confirmed
Stock Movement	Stores every stock increase or decrease.	product, shop, type, quantity, source record, before and after quantity	Confirmed
Purchase	Stores purchase header.	supplier, shop, date, total, payment status, user	Confirmed
Purchase Item	Stores purchased products.	purchase id, product, quantity, unit cost, line total	Confirmed
Sale	Stores sale header.	shop, customer, date, total, payment method, payment status, user	Confirmed
Sale Item	Stores sold products.	sale id, product, quantity, unit price, cost basis, line total	Confirmed
Expense	Stores business expenses.	shop, category, amount, date, user, reason	Confirmed
Capital Entry	Stores capital injections and withdrawals.	shop or business, type, amount, date, reason	Confirmed
Debt	Stores customer or supplier debt.	party, type, original amount, balance, due date, status	Confirmed
Debt Payment	Stores repayments.	debt id, amount, payment date, method, user	Confirmed
Daily Session	Stores daily opening and closing.	shop, business date, opening data, closing data, status	Confirmed
Profit Allocation	Stores 60/40 allocation outputs.	period, profit amount, 60 amount, 40 amount, destination labels	Confirmed
Notification	Stores system alerts.	type, recipient, message, read status, related record	Confirmed
Audit Log	Stores critical actions.	actor, action, entity, record id, before value, after value, timestamp	Confirmed
Sync Queue	Tracks offline mobile changes.	local id, entity, payload, sync status, retries, error	Confirmed

Important Data Rules

Data Rule	Status
Every transaction should belong to a shop and a user.	Confirmed
Soft deletion or deactivation should be used for business records that have transaction history.	Proposed
Stock movement should be append-only so stock history remains auditable.	Confirmed
Audit logs should not be editable by normal users.	Confirmed
Profit records should store the formula version or configuration used to calculate them.	Proposed
60/40 allocation records should store the source profit period and approved profit amount.	Confirmed
Mobile sync records should preserve local creation time and server synchronization time.	Confirmed

Accounting Decisions Required Before Final Schema

Decision	Why It Matters	Status
Profit formula	Determines whether profit means gross profit, net profit, cash profit, or another business-specific measure.	Pending Decision
Stock cost method	Determines cost of goods sold and profit for products bought at different prices.	Pending Decision
Expense treatment	Determines which expenses reduce profit and which are tracked separately.	Pending Decision
Debt categories	Determines whether customer debt, supplier debt, owner debt, and business debt are separate modules or one configurable module.	Pending Decision
60/40 destination labels	Determines what the 60 percent and 40 percent amounts represent in reports.	Pending Decision
Allocation timing	Determines whether 60/40 is applied daily, weekly, monthly, or after owner approval.	Pending Decision
Tax or VAT handling	Determines invoice fields, tax reports, and profit calculation.	Pending Decision